

Background

Concrete is one of the most commonly used construction materials in the world, forming the foundation of buildings, bridges, and roadways. Despite its widespread use, concrete is prone to cracking over time due to weathering, mechanical stress, and internal pressure. These cracks allow water and chemicals to penetrate the material, leading to corrosion of steel reinforcement and a gradual loss of structural integrity. As a result, concrete deterioration presents major safety concerns and contributes to rising maintenance costs.



Figure 1: Map of United States showing 37 states with increase roads in poor condition(2017)

Recent research has introduced self-healing concrete as a potential solution to these challenges. One promising method involves incorporating the bacteria *Bacillus megaterium* into concrete mixtures. These bacteria remain dormant until cracks form and moisture enters the concrete. Once activated, they produce calcium carbonate, which fills and seals cracks naturally. This project investigates whether concrete containing *Bacillus megaterium* demonstrates improved strength and self-repair capabilities compared to traditional concrete.



Figure 2: Concrete before and after bacterial healing agent activation

Purpose

The purpose of this experiment is to determine whether concrete containing *Bacillus megaterium* can heal cracks autonomously and achieve greater strength compared to traditional concrete.

Investigating Bacteria Incorporation into Concrete as a Potential Self-Healing Mechanism for Crack Restoration

Methods

Bacterial Cultivation:

A sample of *Bacillus megaterium* was introduced into a nutrient broth and placed in an oscillating incubator to provide consistent heat and agitation for optimal growth. The culture was monitored over time for signs of bacterial growth while maintaining sterile conditions to prevent contamination.

Mixing and Casting:

Square and cylindrical molds were constructed using wood and PVC materials. Two concrete mixes were prepared with water and calcium acetate nutrient. Control samples were poured first, followed by bacteria-infused concrete, which was then cast into molds and cured.

Cracking & Testing:

After curing, samples underwent initial compressive strength testing using a hydraulic press. Specimens were cracked without full failure and exposed to moisture and heat to promote healing. After 56 days, samples were retested and crack repair was evaluated.



Figure 3: Bacteria cultivation in rotating incubator

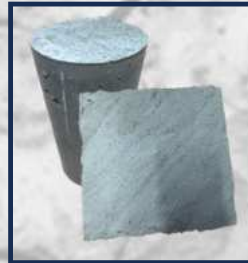


Figure 4: Concrete slab molds constructed by students



Figure 5: Mixing of bacterial concrete solutions



Figure 5: Hydraulic press used for compression testing

Results

Observed Force by Mold Shape (lbf)	Portland Cement	CSA Concrete
Cylinder	11,500	11,000
Slab	8,700	7,900
Cylinder + Bacteria	13,900	13,200
Slab + Bacteria	9,300	8,200

Figure 7: Chart displaying observed force on samples

Chart providing raw data readings from strength testing in the hydraulic press.

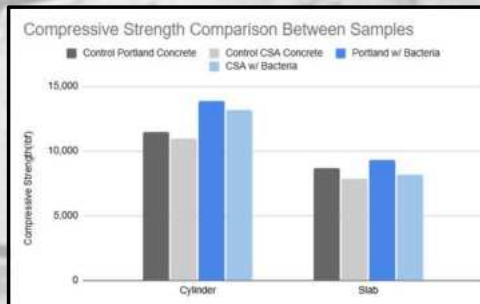


Figure 8: Compressive strength comparison between samples

Graph representing comparison strength testing for bacterial samples against non-bacterial samples

Conclusion

This study investigated whether concrete infused with *Bacillus megaterium* could autonomously repair cracks and improve durability compared to traditional concrete. Based on the results, self-healing remains inconclusive, as measurable crack repair after moisture exposure was not observed. While some visual changes suggested possible mineral activity, there was not enough consistent evidence to confirm effective self-healing across all samples. However, compressive strength data shows a clear increase for samples containing bacteria, with bacterial concrete up to 20% stronger than those without. These results demonstrate that bacterial concrete outperformed conventional concrete in load-bearing capacity. This increase is likely due to calcium carbonate precipitation produced by the bacteria, which can fill microvoids and densify the internal structure. Even without confirmed self-healing, this reinforcement may help reduce crack propagation and improve durability. Overall, the findings suggest bacterial concrete has strong potential as a more durable alternative, though further testing is needed to verify its self-healing capabilities.

Further research

Future research should focus on increasing sample size and completing compressive strength testing to confirm reliability. Additional studies could test different bacterial concentrations and nutrient sources (such as calcium lactate vs. calcium acetate) to determine the most effective mix. Long-term durability testing under real-world conditions, such as freeze-thaw cycles, heavy loads, and chemical exposure, would better assess infrastructure performance. Research could also explore scaling production and evaluating cost efficiency compared to traditional repair methods.

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